a. Height of collimation method

Procedure:

1. In this method, height of collimation (HC) is calculated for each setting of instrument

   Height of collimation (HC)=RL of BM +Back sight
2. The reduced levels of intermediate points (IS) and foresight sight points (FS) are calculated by using the formula.

   RL = HC – IS (or)

   HC – FS
3. When the instrument is shifted to a new position(C.P), a new height of collimation level is setup and the steps (1) and (2) are repeated

   Note:

   HC=Height of collimation level

   BS = Back sight reading

   IS = Intermediate sight reading

   FS = Foresight reading

   RL=Reduced Level

   BM = Bench mark

   C.P= Change Point

b. Rise and fall method

Procedure:

1. In this method the difference of level between consecutive points is found by comparing staff readings Rise indicates (+) and fall indicates (-).

   i.e., BS – IS (Or) IS – IS (or)
   I.S – F.S
2. RL of any point = (RL of preceding point) + Rise (or) - fall



Digital Level

- They are not popular instead auto levels are more extensively used.
- The Trimble DiNi Digital level: Determine accurate height information 60% faster than with automatic levelling.
- Eliminate errors and reduce rework with digital readings.
- Transfer data to the office easily.
- Measure to a field of just 30 cm.



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